Lctr 21: DT System Analysis and the Unilateral z-Transform

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Spring 2026

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Focus: Sections 10.7–10.9 (Pages 774–796)

21.1 The Big Picture

Why This Matters

This lecture completes the z-transform trilogy by connecting the machinery to DT system analysis. The goals mirror Lecture 18 exactly:
Part I: Given a difference equation, find the system function $H(z)$, determine stability and causality, compute the output.
Part II: Use the unilateral z-transform to solve difference equations with nonzero initial conditions (the DT analog of solving IVPs with the unilateral Laplace transform).
After this lecture, you have a complete toolkit for both CT and DT system analysis.

Roadmap:

1. The system function $H(z)$ from difference equations.
2. Causality and stability in the z-domain.
3. Complete system analysis example.
4. Block diagram representations.
5. The unilateral z-transform and its time-shift property.
6. Solving difference equations with initial conditions.

Part I: System Analysis with $H(z)$

21.2 The System Function $H(z)$

A causal LTI DT system described by a linear constant-coefficient difference equation:

$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k]$$

Taking the bilateral z-transform (zero initial conditions): each delay $y[n-k]$ becomes $z^{-k}Y(z)$, each $x[n-k]$ becomes $z^{-k}X(z)$:

$$\left(\sum_{k=0}^{N} a_k\, z^{-k}\right) Y(z) = \left(\sum_{k=0}^{M} b_k\, z^{-k}\right) X(z)$$

$$\boxed{H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k\, z^{-k}}{\sum_{k=0}^{N} a_k\, z^{-k}} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}}}$$

Key Insight

The z-transform converts the difference equation into an algebraic equation: each unit delay $n \to n-1$ becomes multiplication by z^{-1}. This is why z^{-1} is called the "delay operator" in digital signal processing.

From Difference Equation to $H(z)$

System: $y[n] - 0.8\,y[n-1] + 0.15\,y[n-2] = x[n] + 2x[n-1]$
Step 1: Replace delays with z^{-k}:

$$(1 - 0.8z^{-1} + 0.15z^{-2})Y(z) = (1 + 2z^{-1})X(z)$$

Step 2: Form $H(z)$:

$$H(z) = \frac{1 + 2z^{-1}}{1 - 0.8z^{-1} + 0.15z^{-2}}$$

Step 3: Factor the denominator. We need roots of $1 - 0.8z^{-1} + 0.15z^{-2} = 0$, i.e., $z^2 - 0.8z + 0.15 = 0$:

$$z = \frac{0.8 \pm \sqrt{0.64 - 0.60}}{2} = \frac{0.8 \pm 0.2}{2}$$

$z_1 = 0.5$, $z_2 = 0.3$.
So:

$$\boxed{H(z) = \frac{1 + 2z^{-1}}{(1 - 0.5z^{-1})(1 - 0.3z^{-1})}}$$

Poles: $z = 0.5, 0.3$ (both inside the unit circle). Zero: from $1 + 2z^{-1} = 0$, $z = -2$.

21.3 Causality and Stability

Causality Criterion

A DT LTI system is **causal** iff $h[n] = 0$ for $n < 0$, which requires the ROC to be the **exterior of a circle**: $|z| > r_{\max}$ (outside the outermost pole).
For rational $H(z)$: causality also requires $M \leq N$ (numerator degree $\leq$ denominator degree in z^{-1}), or equivalently $H(z) \to$ finite as $z \to \infty$.

Stability Criterion

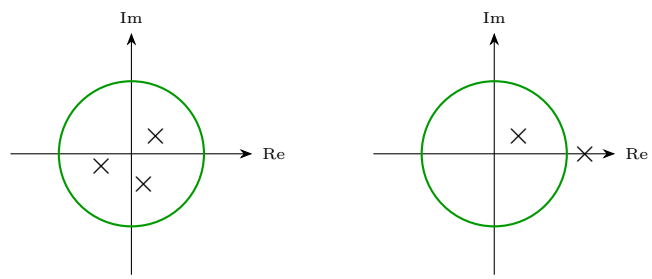
A DT LTI system is **BIBO stable** iff $\sum |h[n]| < \infty$, which requires the ROC to **include the unit circle** ($|z| = 1$).

The Golden Rule for Causal DT Systems

A causal LTI DT system with rational $H(z)$ is BIBO stable **if and only if all poles lie strictly inside the unit circle**:

$$\text{Causal} + \text{Stable} \iff |p_i| < 1 \;\; \forall \text{ poles } p_i$$

Why: For a causal system, ROC is $|z| > |p_{\max}|$. For the unit circle to be in the ROC, we need $|p_{\max}| < 1$.



Causal + Stable
All poles inside $|z| = 1$

Causal + Unstable
A pole outside $|z| = 1$

Stability Classification

$H(z)$	**Poles**	$\lvert p_i \rvert$	**Causal + Stable?**
$\dfrac{1}{1-0.5z^{-1}}$	$z = 0.5$	$0.5 < 1$	**Yes**
$\dfrac{1}{1-1.2z^{-1}}$	$z = 1.2$	$1.2 > 1$	**No (unstable)**
$\dfrac{1}{1-z^{-1}}$	$z = 1$	$1 = 1$	**Marginally stable**
$\dfrac{1}{(1-0.3z^{-1})(1+0.7z^{-1})}$	$z = 0.3, -0.7$	$0.3, 0.7 < 1$	**Yes**

21.4 Complete System Analysis Pipeline

Full Pipeline: Difference Equation to Output

System: $y[n] - 0.5\,y[n-1] = x[n]$, **Input:** $x[n] = (0.8)^n u[n]$.
Step 1: Find $H(z)$.

$$(1 - 0.5z^{-1})Y = X \;\;\Rightarrow\;\; H(z) = \frac{1}{1-0.5z^{-1}}\,, \quad |z| > 0.5$$

Pole at $z = 0.5$ (inside unit circle $\Rightarrow$ causal and stable). ✓
Step 2: Find $X(z)$.

$$X(z) = \frac{1}{1-0.8z^{-1}}\,, \quad |z| > 0.8$$

Step 3: Find $Y(z) = H(z) \cdot X(z)$.

$$Y(z) = \frac{1}{(1-0.5z^{-1})(1-0.8z^{-1})}, \quad |z| > 0.8$$

Step 4: Partial fractions.

$$\frac{1}{(1-0.5z^{-1})(1-0.8z^{-1})} = \frac{A}{1-0.5z^{-1}} + \frac{B}{1-0.8z^{-1}}$$

Set $z^{-1} = 2$: $1 = A(1-1.6) \Rightarrow A = -5/3$.
Set $z^{-1} = 1.25$: $1 = B(1-0.625) \Rightarrow B = 8/3$.
Step 5: Invert. Both poles inside ROC $\Rightarrow$ both right-sided:

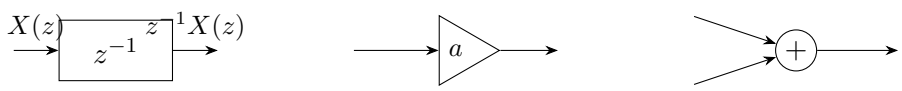
$$\boxed{y[n] = \left[-\frac{5}{3}(0.5)^n + \frac{8}{3}(0.8)^n\right] u[n]}$$

Step 6: Verify.

- $y[0] = -5/3 + 8/3 = 3/3 = 1 = x[0]$ ✓ (from the difference equation with $y[-1] = 0$)
- $y[1] = -5/3 \cdot 0.5 + 8/3 \cdot 0.8 = -5/6 + 6.4/3 = -5/6 + 32/15$. From the ODE: $y[1] = 0.5y[0] + x[1] = 0.5 + 0.8 = 1.3$. Check: $-5/6 + 32/15 = -25/30 + 64/30 = 39/30 = 1.3$ ✓

21.5 Block Diagram Representations

In the z-domain, the three basic building blocks are:



Unit Delay Gain Adder

Every rational $H(z)$ can be implemented as a combination of delays (z^{-1}), multipliers (gains), and adders. This is the basis of **digital filter design**.

Key Insight

The block z^{-1} represents storing one sample in memory. A difference equation of order N requires exactly N memory elements. In software, each z^{-1} is one element of a circular buffer; in hardware, it is one register or latch. This direct correspondence between the z-domain and digital implementation is why the z-transform is the language of DSP.

Part II: The Unilateral z-Transform

21.6 Definition

The Unilateral z-Transform

$$\boxed{\mathcal{X}(z) = \sum_{n=0}^{\infty} x[n]\, z^{-n}}$$

The sum starts at $n = 0$, not $n = -\infty$. For causal signals ($x[n] = 0$ for $n < 0$), the unilateral and bilateral transforms are identical.

21.7 The Key Property: Time Shifting

The bilateral z-transform gives: $x[n-1] \mathcal{Z} z^{-1} X(z)$.

The unilateral z-transform gives:

$$\boxed{x[n-1] \;\; \mathcal{Z}_u \;\; z^{-1}\mathcal{X}(z) + x[-1]}$$

$$\boxed{x[n-2] \;\; \mathcal{Z}_u \;\; z^{-2}\mathcal{X}(z) + x[-2] + x[-1]\, z^{-1}}$$

General pattern for delay by m:

$$x[n-m] \;\; \mathcal{Z}_u \;\; z^{-m}\mathcal{X}(z) + x[-m] + x[-m+1]\, z^{-1} + \cdots + x[-1]\, z^{-(m-1)}$$

Key Insight

The initial conditions $x[-1], x[-2], \ldots$ appear explicitly, just as $x(0^-), x'(0^-)$ appeared in the unilateral Laplace differentiation property. This is the mechanism for incorporating nonzero initial conditions into difference equation solutions.

21.8 Solving Difference Equations with Initial Conditions

First-Order Difference Equation with IC

Solve: $y[n] - 0.6\, y[n-1] = (0.5)^n u[n], \quad y[-1] = 4.$

Step 1: Unilateral z-transform.

For the left side:

$$Y(z) - 0.6\big[z^{-1}Y(z) + y[-1]\big] = Y(z) - 0.6z^{-1}Y(z) - 0.6 \cdot 4$$

$$= (1 - 0.6z^{-1})Y(z) - 2.4$$

For the right side: $\dfrac{1}{1 - 0.5z^{-1}}$.

Step 2: Solve for $Y(z)$.

$$(1 - 0.6z^{-1})Y(z) = \frac{1}{1 - 0.5z^{-1}} + 2.4$$

$$Y(z) = \frac{1}{(1 - 0.5z^{-1})(1 - 0.6z^{-1})} + \frac{2.4}{1 - 0.6z^{-1}}$$

Step 3: Partial fractions on the first term.

$$\frac{1}{(1 - 0.5z^{-1})(1 - 0.6z^{-1})} = \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.6z^{-1}}$$

Set $z^{-1} = 2$: $1 = A(1 - 1.2) \Rightarrow A = -5$.
Set $z^{-1} = 5/3$: $1 = B(1 - 5/6) = B/6 \Rightarrow B = 6$.
So:

$$Y(z) = \frac{-5}{1 - 0.5z^{-1}} + \frac{6}{1 - 0.6z^{-1}} + \frac{2.4}{1 - 0.6z^{-1}} = \frac{-5}{1 - 0.5z^{-1}} + \frac{8.4}{1 - 0.6z^{-1}}$$

Step 4: Invert.

$$\boxed{y[n] = \big[-5 \cdot (0.5)^n + 8.4 \cdot (0.6)^n\big]\, u[n]}$$

Step 5: Verify.
$y[0] = -5 + 8.4 = 3.4$. From the ODE: $y[0] = 0.6 \cdot y[-1] + x[0] = 0.6 \cdot 4 + 1 = 3.4$ ✓.
$y[1] = -5(0.5) + 8.4(0.6) = -2.5 + 5.04 = 2.54$. From the ODE: $y[1] = 0.6 \cdot y[0] + x[1] = 0.6 \cdot 3.4 + 0.5 = 2.54$ ✓.

Second-Order

Solve: $y[n] - 0.7\,y[n-1] + 0.1\,y[n-2] = 0, \quad y[-1] = 1,\ y[-2] = 0.$

Step 1: Unilateral z-transform.

$$\mathcal{Z}_u\{y[n-1]\} = z^{-1}Y(z) + y[-1] = z^{-1}Y + 1$$
$$\mathcal{Z}_u\{y[n-2]\} = z^{-2}Y(z) + y[-2] + y[-1]z^{-1} = z^{-2}Y + 0 + z^{-1}$$

Substitute:

$$Y - 0.7(z^{-1}Y + 1) + 0.1(z^{-2}Y + z^{-1}) = 0$$
$$(1 - 0.7z^{-1} + 0.1z^{-2})Y = 0.7 - 0.1z^{-1}$$

Step 2: Factor. $z^2 - 0.7z + 0.1 = 0 \Rightarrow z = \frac{0.7 \pm \sqrt{0.49 - 0.4}}{2} = \frac{0.7 \pm 0.3}{2}$. Poles: $z = 0.5$ and $z = 0.2$.

$$Y(z) = \frac{0.7 - 0.1z^{-1}}{(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

Step 3: Partial fractions.

$$\frac{0.7 - 0.1z^{-1}}{(1 - 0.5z^{-1})(1 - 0.2z^{-1})} = \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.2z^{-1}}$$

Set $z^{-1} = 2$: $0.7 - 0.2 = A(1 - 0.4) \Rightarrow 0.5 = 0.6A \Rightarrow A = 5/6$.
Set $z^{-1} = 5$: $0.7 - 0.5 = B(1 - 2.5) \Rightarrow 0.2 = -1.5B \Rightarrow B = -2/15$.
Step 4: Invert.

$$\boxed{y[n] = \left[\frac{5}{6} \cdot (0.5)^n - \frac{2}{15} \cdot (0.2)^n\right] u[n]}$$

This is a pure **zero-input response**: there is no external input, so the output is entirely due to the initial conditions. Both modes decay because both poles are inside the unit circle.
Verify: $y[0] = 5/6 - 2/15 = 25/30 - 4/30 = 21/30 = 0.7$. From ODE: $y[0] = 0.7 \cdot y[-1] - 0.1 \cdot y[-2] = 0.7 \cdot 1 - 0 = 0.7$ ✓.

21.9 ZSR + ZIR Decomposition

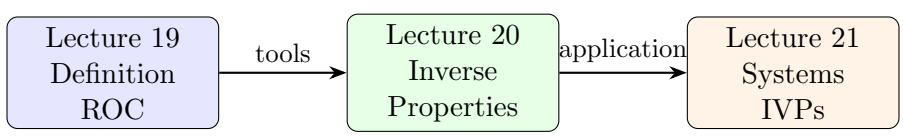
Just as in CT (Lecture 18):

$$y[n] = \underbrace{y_{\text{ZS}}[n]}_{\text{zero-state}} + \underbrace{y_{\text{ZI}}[n]}_{\text{zero-input}}$$

$y_{\text{ZS}}[n]$: output due to input alone, with all initial conditions set to zero.

$y_{\text{ZI}}[n]$: output due to initial conditions alone, with zero input.

21.10 Summary of the Three-Lecture z-Transform Block



	Laplace (Lctrs 16–18)	**z-Transform (Lctrs 19–21)**
Converts	differential eqs $\to$ algebra	difference eqs $\to$ algebra
Delay operator	e^{-st_0}	z^{-1}
Causal + Stable	poles in LHP ($\text{Re}\{p\} < 0$)	poles inside $\lvert z\rvert = 1$ ($\lvert p\rvert < 1$)
Unilateral IC property	$y'(t) \to sY - y(0^-)$	$y[n-1] \to z^{-1}Y + y[-1]$
FT exists	$j\omega$-axis in ROC	unit circle in ROC

Key takeaways:

1. $H(z)$ from a difference equation: replace each delay $n - k$ with z^{-k} and solve for Y/X.
2. Causal + stable $\Leftrightarrow$ all poles strictly inside the unit circle.
3. Block diagrams use z^{-1} as the basic delay element; this maps directly to digital hardware/software.

4. The unilateral z-transform handles initial conditions via the modified time-shift property: $y[n-1] \to z^{-1}Y + y[-1]$.

5. Total response = ZSR + ZIR.

21.11 Common Mistakes to Avoid

1. **Confusing "inside unit circle" with "LHP":** In the z-domain, stability requires poles *inside* the unit circle (not in the left half-plane). A pole at $z = -0.9$ is stable (inside the circle) even though it is on the negative real axis.

2. **Getting the unilateral shift property wrong:** For $y[n-1]$: the IC term is $+y[-1]$, **not** $-y[-1]$. Compare with Laplace: $y'(t) \to sY - y(0^-)$ has a minus sign, but the z-domain version has a plus sign.

3. **Forgetting that z^{-1} means delay, not advance:** $x[n-1] \leftrightarrow z^{-1}X(z)$ is a delay. $x[n+1] \leftrightarrow zX(z) - zx[0]$ is an advance (unilateral). Do not confuse the two.

4. **Misidentifying pole locations from the z^{-1} form:** In $H(z) = \frac{1}{1-az^{-1}}$, the pole is at $z = a$, not $z = 1/a$ or $z = -a$.

5. **Checking stability by testing $a < 1$ instead of $|a| < 1$:** A pole at $z = -0.9$ has $|z| = 0.9 < 1$, so it is stable. But $a = -0.9 < 1$ is not the right check; $|a| < 1$ is.

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